NOTES 4.5 – Quadratics: Standard Form to Vertex Form

Standard Form:

$ax^2 + bx + c = 0$

Vertex Form:

$a(x - h)^2 + k$

$h = x \text{ value of vertex}$

$k = y \text{ value of vertex}$

$y = 8x^2 - 16x + 27$ $\Rightarrow$ $y = 8(x-1)^2 + 19$

$y = 5x^2 - 40x + 67$ $\Rightarrow$ $y = 5(x-4)^2 - 13$

Advantages of Standard Form:

- *Can read y-int (x = 0) from Equation*
- *Can find x-ints (y = 0) by…*
  - *Quadratic Formula, Factoring, Solving Radicals, & Completing the Square*

Advantages of Vertex Form:

- *Can read Vertex from Equation*

How do you convert from Standard to Vertex Form?

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- Find h (use "first part" of Quad. FM.)
- Find k (plug h into EQ and solve)

$\underline{h = x \text{ value of vertex}}$

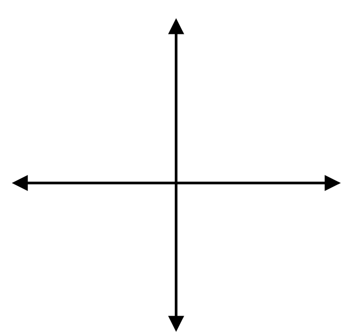
$\underline{k = y \text{ value of vertex}}$

$$h = \frac{-b}{2a} \Longleftrightarrow k = a(h)^2 + b(h) + c$$

Examples:

a) $y = 8x^2 - 16x + 27$

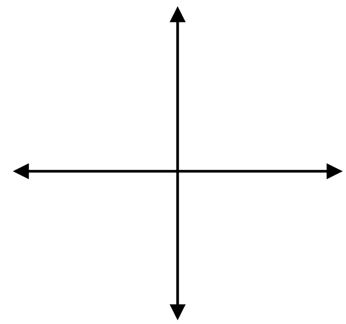
Sketch Parabola & label y –int and vertex



Vertex________________Equation (vertex form)______________________________

b) $y = -x^2 + 6x - 5$

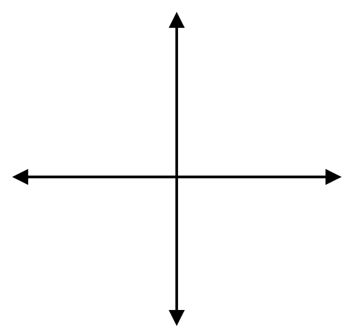
Sketch Parabola & label y –int and vertex



Vertex_________________Equation (vertex form)_______________________________

c) $y = 7x^2 + 28x + 19$

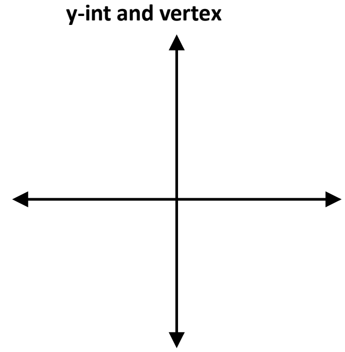
Sketch Parabola & label y –int and vertex



Vertex_________________Equation (vertex form)_______________________________

d) $y = -2x^2 - 24x - 75$

Sketch Parabola & label y-int and vertex



Vertex_________________Equation (vertex form)_______________________________